

5.

Exercicio 5.1

a)

```
f[x_, y_, z_] = x^2 + 2 y^2 + 3 z^2; ponto = {1, 1, 1};

f @@ ponto

6

Gradf[x_, y_, z_] = Grad[f[x, y, z], {x, y, z}]

{2 x, 4 y, 6 z}
```

Reta normal

y = -1+2 x
z = -2+3 x

Plano tangente

z = 1/3 (6 - x - 2 y)

b)

```
f[x_, y_, z_] = x y z^2; ponto = {1, 1, 1};

f @@ ponto

1

Gradf[x_, y_, z_] = Grad[f[x, y, z], {x, y, z}]

{y z^2, x z^2, 2 x y z}
```

Reta normal

y = x
z = -1+2 x

Plano tangente

z = 1/2 (4 - x - y)

c)

```
f[x_, y_, z_] = x^2 + 3 y^3 + Sin[x y] - z; ponto = {1, 0, 1};

f @@ ponto

0

Gradf[x_, y_, z_] = Grad[f[x, y, z], {x, y, z}]

{2 x + y Cos[x y], 9 y^2 + x Cos[x y], -1}
```

Reta normal

y = 1/2 (-1 + x)
z = 3/2 - x/2

Plano tangente

z = -1 + 2 x + y

d)

```
f[x_, y_, z_] = Exp[x y z]; ponto = {1, 1, 0};

f @@ ponto

1

Gradf[x_, y_, z_] = Grad[f[x, y, z], {x, y, z}]

{e^{x y z} y z, e^{x y z} x z, e^{x y z} x y}
```

Reta normal

x = 1

Plano tangente

z = 0

Exercício 5.2

a)

f[x_, y_, z_] = x^3 + x y z; ponto = {2, 2, 1};

f@@ponto

12

Grad[f[x_, y_, z_] = Grad[f[x, y, z], {x, y, z}]

{3 x^2 + y z, x z, x y}

Reta normal

y = $\frac{12 + x}{7}$

z = $\frac{3}{7} + \frac{2 x}{7}$

Plano tangente

z = $\frac{1}{2} (18 - 7 x - y)$

b)

Não

Exercicio 5.3

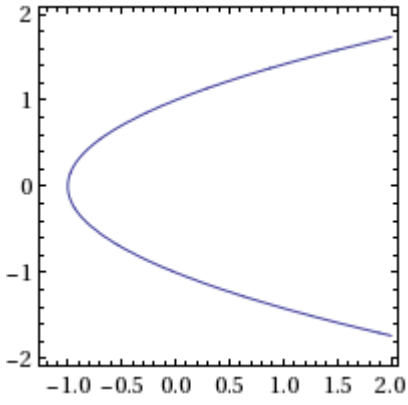
a)

f[x_, y_] = x - y^2; A = {-1, 0};

f@@A

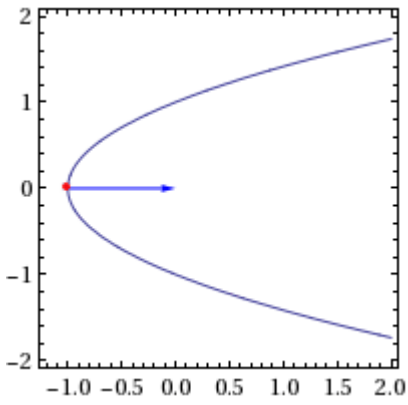
-1

x = -1 + y^2



b)

{1, -2 y}



c)

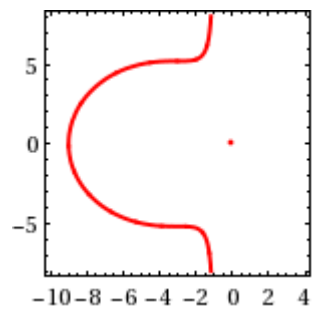
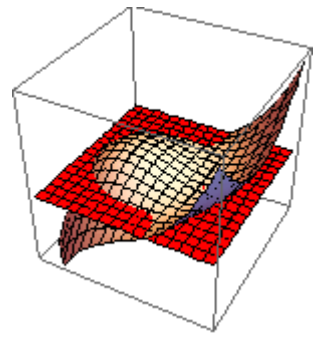
z = x

Exercicio 5.4

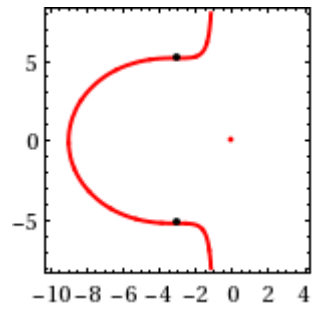
f[x_, y_] = x (x^2 + y^2) + 9 x^2 + y^2;

Grad[f[x, y], {x, y}]

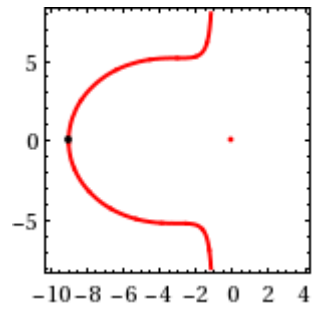
{18 x + 3 x^2 + y^2, 2 y + 2 x y}



$\left\{ \left(-3, -3\sqrt{3}\right), \left(-3, 3\sqrt{3}\right) \right\}$



$\{(-9, 0)\}$



Exercício 5.5

$\left\{ (0, 1), \left(\frac{2}{3}, -\frac{1}{3}\right) \right\}$

Exercício 5.6

$\left\{ \left(\frac{2}{3}, -\frac{4}{3}\right), (2, 0) \right\}$

Exercício 5.7

$f(x, y, z) = x^2 + y^2 + z^2;$

$z = \frac{5+y}{2}$

c

$x = \frac{5-y}{2}$

Exercício 5.8

$\forall f(P) \cdot e_1 < 0; \forall f(Q) \cdot e_2 > 0$